

08.1

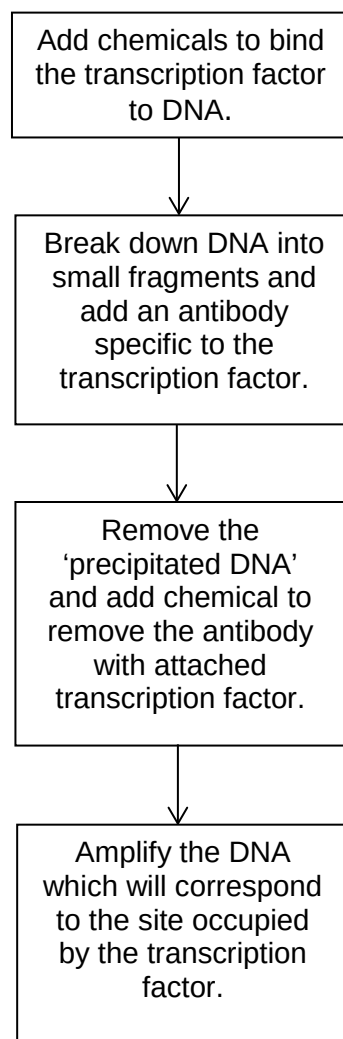
What is meant by a genome?

[1 mark]

out

Chromatin immunoprecipitation is one method to determine where a transcription factor binds to DNA. The principle behind this procedure is shown in **Figure 8**.

Figure 8



08.2

Explain why the antibody binds to the transcription factor.

[2 marks]

08.3

Use **Figure 8** to explain what 'precipitated DNA' consists of.

[1 mark]

Soybeans are used in a number of processed foods. However, soybeans contain a protein known as P34 that causes an allergic response in some people. Scientists have created transgenic soybeans that produce single-stranded cDNA, which prevents transcription of the *P34* gene. They used recombinant plasmids as vectors to transform soybean cells. After they had screened these cells for production of the P34 protein, they cultured the transformed cells to form soybean plants.

08.4

Suggest how single-stranded cDNA could prevent transcription of the *P34* gene.

[1 mark]

Question 8 continues on the next page

0 8 . 5

Describe the roles of **two** named types of enzymes used to insert DNA fragments into plasmids.

[2 marks]

Type of enzyme _____

Role _____

Type of enzyme _____

Role _____

0 8 . 6

The soybean cells were screened for the presence of the P34 protein. This process involved the use of gel electrophoresis to separate proteins extracted from soybean cells.

Suggest **two** features of the structure of different proteins that enable them to be separated by gel electrophoresis.

[2 marks]

1 _____

2 _____
